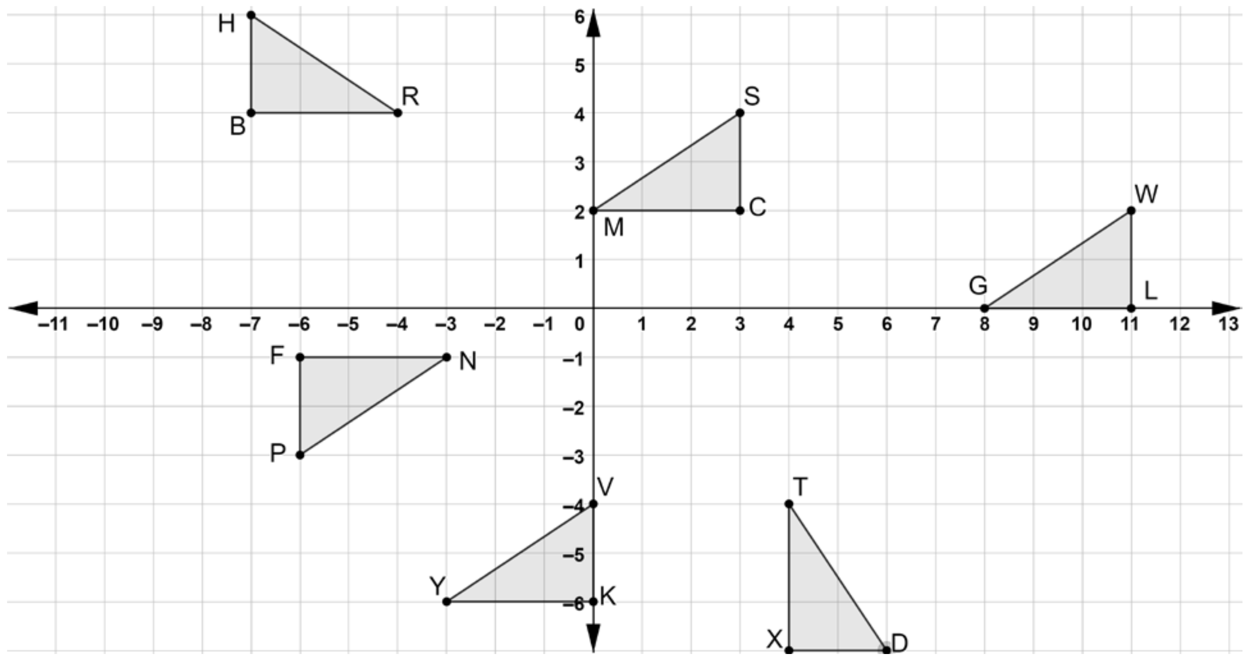
Geometry
Unit 4
Lesson 8 Practice

Name_Answer Key________
Date___________________
Period__________________

Six triangles are shown on the coordinate grid.



Give a written description of the transformations that lead to the given congruence statement where $\triangle FNP$ is the preimage.

1. $\triangle FNP \cong \triangle CMS$

   Rotate 180° about the origin, translate left 3 units, and up 1 unit.

2. $\triangle FNP \cong \triangle XTD$

   Rotate 90° counterclockwise about the origin, translate right 3 units, and down 1 unit.

3. $\triangle FNP \cong \triangle KYV$

   Rotate 180° about the origin, translate left 6 units, and down 7 units.

4. $\triangle FNP \cong \triangle BRH$

   Reflect over x-axis, translate left 1 unit, and up 3 units.

5. $\triangle FNP \cong \triangle LGW$

   Rotate 180° about the origin, translate right 5 units, and down 1 unit.

Give an algebraic description of the transformations that lead to the given congruence statement where $\triangle FNP$ is the preimage.

6. $\triangle FNP \cong \triangle CMS$

$(x, y) \rightarrow (-x, -y)$

$(x, y) \rightarrow (x - 3, y + 1)$

7. $\triangle FNP \cong \triangle XTD$

$(x, y) \rightarrow (-y, x)$

$(x, y) \rightarrow (x + 3, y - 1)$

8. $\triangle FNP \cong \triangle KYV$

$(x, y) \rightarrow (-x, -y)$

$(x, y) \rightarrow (x - 6, y - 7)$

9. $\triangle FNP \cong \triangle BRH$

$(x , y) \rightarrow (x, -y)$

$(x, y) \rightarrow (x - 1, y + 3)$

10. $\triangle FNP \cong \triangle LGW$

$(x, y) \rightarrow (-x, -y)$

$(x, y) \rightarrow (x + 5, y - 1)$